



BITS Pilani

Pilani | Dubai | Goa | Hyderabad | Mumbai

**WORK INTEGRATED
LEARNING PROGRAMMES**

46
Years
of Delivering
Excellence

M.Tech.

Artificial Intelligence and Machine Learning

for working professionals

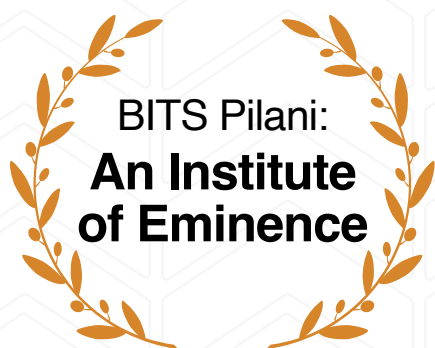
Learn. Work. Grow.



Programme Brochure

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BITS Pilani:

Since 1964, BITS Pilani has been a pioneer in higher education, recognised as an Institute of Eminence by the UGC and Ministry of Education. Known for academic excellence, it has grown into a globally respected institution and was the first Indian university to establish an international campus in Dubai.

Welcome to BITS Pilani's Work Integrated Learning Programmes (WILP)

At BITS Pilani WILP, we blend education with professional growth through tailored programmes aligned with both business and learner goals. Professionals gain new skills, adopt best practices, and make a meaningful impact, while organisations benefit from increased productivity, retention, and a motivated workforce. Our UGC-approved programmes offer real-world, practical knowledge, ensuring lasting success.

46

Years of legacy

1,36,904

Working Professionals
Graduated

353

CEOs and
Founders

4,997

Global Heads,
Presidents & Directors

367

Organisations Trust Us
As Learning Partner

50,117

Working Professionals
Currently Enrolled



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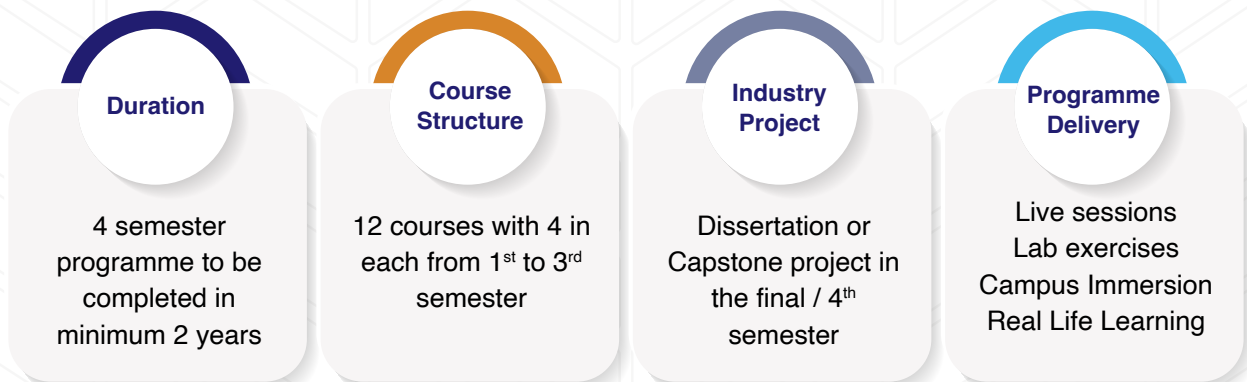
M.Tech. Artificial Intelligence and Machine Learning

Programme Introduction

With a comprehensive curriculum, hands-on experiential learning through remote and cloud labs, and flexible education methods, this programme prepares professionals for limitless career possibilities in Artificial Intelligence and Machine Learning.

Designed as a four-semester course for IT professionals and software developers, it enables learners to build a future-ready skill set and advance into high-demand roles such as Machine Learning Engineer and AI Scientist — all without pausing their careers.

Programme Structure



Eligibility Criteria

- Employed professionals holding B.E. / B.Tech. with at least 60% aggregate marks and minimum one-year relevant work experience after the completion of the degree are eligible to apply.
- Employed professionals holding MCA / M.Sc. or equivalent with at least 60% aggregate marks with university level mathematics / statistics as mandatory subjects and minimum one-year relevant work experience after the completion of the degree are also eligible to apply.
- Working knowledge of Computing and programming is required.

Who Should Apply

IT and Software professionals working as Software Engineer, Software Developer, Programmer, Software Test Engineer, Support Engineer, Data Analyst, Business Analyst, who wish to transition to roles such as ML Engineers & AI Scientists, etc. should consider applying to this programme.



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M.Tech. Artificial Intelligence and Machine Learning

Programme Objectives

The program focuses on essential skills required for lucrative roles in Data Science, Data Engineering, and Advanced Analytics, covering:



Abundance of user-generated data, easy access to computing and storage in the cloud, open-source libraries and algorithmic advancement have led to the deployment of artificial intelligence and machine learning techniques across industries.



This in turn has fuelled significant job opportunities in the IT products and services sector in India and across the globe.



This program is geared towards the professional development of employees who are working in the area of IT products and services industry or who aspire for a career in the applications of AI and ML techniques in traditional industries.

Career Outcomes

The programme will build you for aspirational career as

- | | | | |
|-------------------|--------------------|------------------|------------------------|
| Software Engineer | Software Developer | Programmer | Software Test Engineer |
| Support Engineer | Data Analyst | Business Analyst | AI Engineer |
| AI Developer | NLP Engineer | AI Architect | AI Product Manager |



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M.Tech. Artificial Intelligence and Machine Learning

Key Benefits of BITS Pilani WILP



Unlock Cutting-Edge Technologies & Industry Practices

Stay ahead of the curve with future-ready tools, methods, and frameworks used by leading professionals worldwide.



Master Hands-On Learning Skills

Apply real world concepts in virtual & remote labs, case studies and simulations that build practical mastery.



Be a Top-Notch Professional in Your Field

Sharpen your expertise, build leadership skills, and accelerate your career growth with industry-relevant learning.



Learn from Best Minds & Industry Experts

Gain insights and mentorship from top faculty and experienced professionals who shape the future of industries.



Grow Without a Career Break

Continue your learning alongside your job, upgrading your skills without taking a career break.



Become Alumni of a Top Institution

Earn credentials from a globally recognised institution and join a powerful, lifelong alumni network

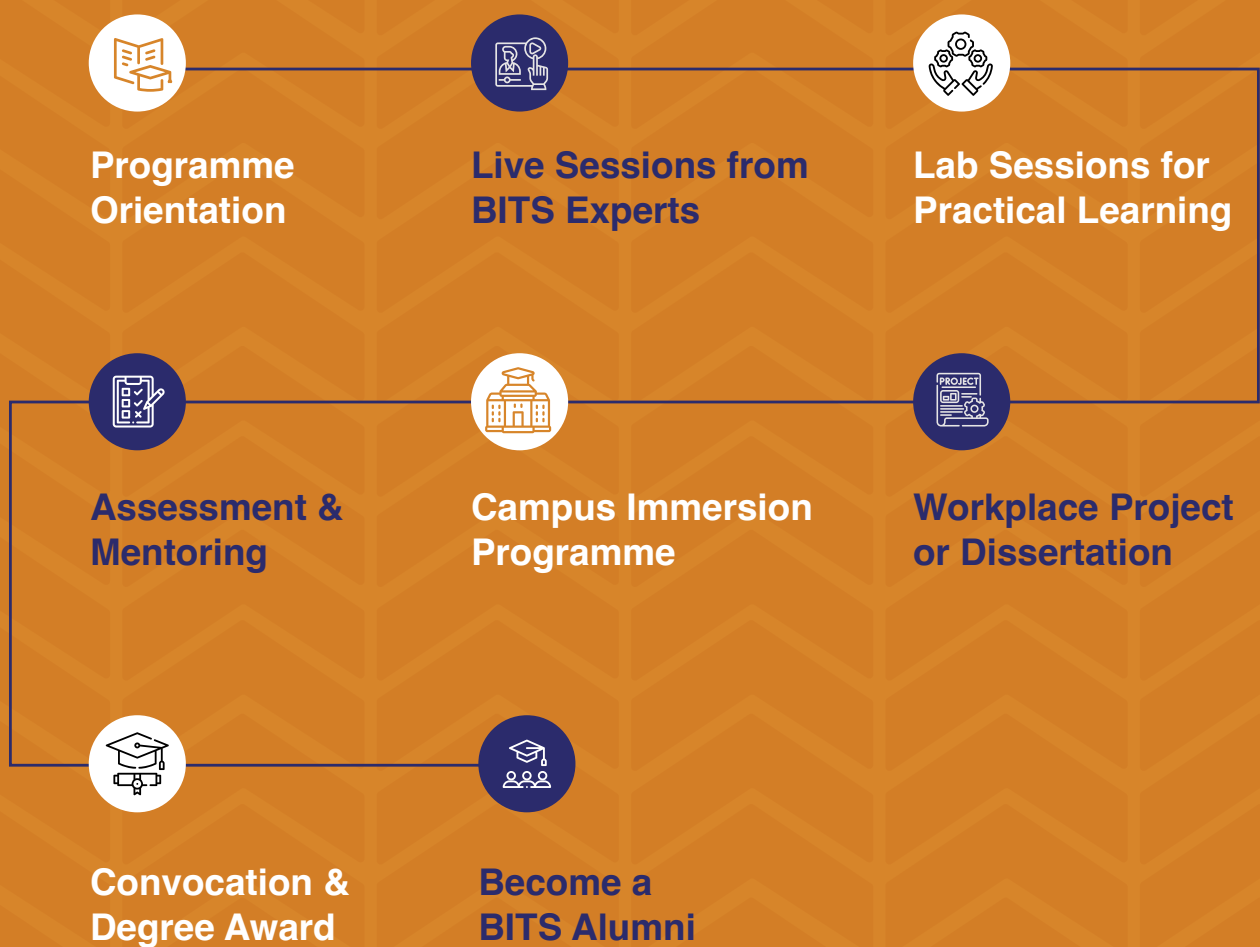


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M.Tech. Artificial Intelligence and Machine Learning

Student Learning Journey

Our programmes are designed to help you grow, excel, and advance in your career. By combining industry-relevant skills with academic excellence, we ensure that your learning journey leads to real, measurable progress. Whether you're looking to boost productivity, stay motivated, or take the next step in your professional path, our courses empower you to achieve more. Invest in yourself today and build a future where you're not just keeping up but leading the way.



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M.Tech. Artificial Intelligence and Machine Learning

Live Interactive and Engaging Learning Methodology



Live Learning

Live sessions for each course

32 hours of live instruction of each course on weekends in an interactive mode.

Faculty-guided | hands-on learning

Solve real life case studies, assignments & more and receive feedback on the same from the faculty members.

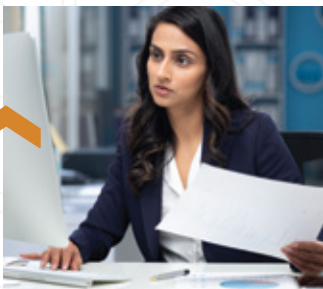
Experiential Learning

Experiential Learning Centres

State-of-the-art centres designed for hands-on, lab-based learning and real-world application. Now live with the Automotive Competency Center in Chennai and Smart Manufacturing Competency Center in Bengaluru, with more locations launching across major cities in 2026.

Campus Immersion Programme

Opportunity to visit the BITS campus to learn, experience labs, network with peers and get mentored by BITS leadership.



Project Based Learning

Final Semester Project

Learners pick project from their workplace or industry dissertation to apply their learning and suggest improvements.

Develop Problem Solving Skills

Identify, analyse, and effectively find solutions utilising critical thinking & analytical reasoning.

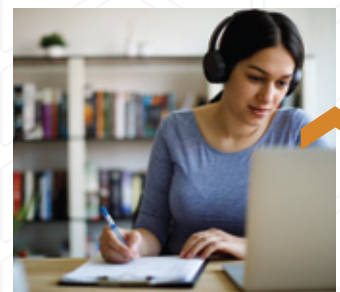
Self-Learning

Flexible learning with access to digital resources

Learners can access e-learning content, recorded sessions, and assignments on our LMS anytime, anywhere.

Library access to books & journals.

Learners get subscription to the world's largest collection of eBooks and other reading resources through OpenAthens.



Flexible Learning

You can take a break to attend personal or professional commitment. Come back & join from where left (Please take advise from your programme co-ordinator on details).



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Labs & Practical Session

Our programmes are designed for real-world impact. With hands-on simulations and remote labs, you don't just learn you apply. Gain practical experience, sharpen problem-solving skills, and master industry-relevant techniques in a dynamic, immersive environment.



Virtual Labs

Software (for design, analysis, modelling, and simulation) hosted on the cloud and accessible online from anywhere.



Remote Labs

Physical lab equipment operated & controlled by students from remote locations.



Simulation, AR / VR & Web Based

Manipulate parameters to understand their impact in a real-world system.

Lab Infrastructure

Access private cloud-based remote labs with specialized AI/ML platforms including TensorFlow, PyTorch, and Keras. Work with high-performance GPU clusters for model training and deep learning frameworks.

Hands-on Experience

Build neural networks, develop computer vision applications, create natural language processing models, and implement machine learning algorithms. Train deep learning models on real datasets and deploy AI solutions using industry-standard tools.



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Tools & Technologies You Will Learn



Campus Immersion Programmes



Learners have the opportunity to visit BITS Pilani campuses for specially curated immersion modules. These sessions offer face-to-face interactions with faculty, hands-on lab experiences, peer networking, and a glimpse into the vibrant campus life.



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Earn a Prestigious Degree and Alumni Status of BITS Pilani



Upon successful completion of the programme, learners are awarded a formal degree during the convocation ceremony, just like on-campus students. This degree comes with the prestigious BITS Pilani alumni status, unlocking lifelong recognition and connection to a global network.

About BITS Alumni Association



Graduates can join BITSAA (BITS Alumni Association), a thriving global community of BITSians. Membership offers access to alumni events, mentorship opportunities, and a range of career and networking benefits. The registration process is simple and can be completed online post-graduation.



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M.Tech. Artificial Intelligence and Machine Learning

Programme Curriculum

First Semester

- Mathematical Foundations of Machine Learning
 - Introduction to Statistical Methods
 - Deep Neural Networks
 - Machine Learning
-

Second Semester

- Artificial and Computational Intelligence
 - Deep Reinforcement Learning
 - Elective - 1
 - Elective - 2
-

Third Semester

- Elective - 3
 - Elective - 4
 - Elective - 5
 - Elective - 6
-

Fourth Semester

- Dissertation
-

Pool of Electives – Deep Learning Specialization

- Unsupervised Deep Learning[#]
- Graph Neural Networks
- Distributed Machine Learning
- ML System Optimization

- Fair, Interpretable, Trustworthy Machine Learning
 - Computational Learning Theory
 - Machine Learning on the Edge
-

[#]At least 3 courses are required including the course marked in

Core Courses

- Mathematical Foundations for Machine Learning
- Introduction to Statistical Methods
- Artificial and Computational Intelligence

- Machine Learning
- Deep Neural Networks
- Deep Reinforcement Learning



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Pool of Electives – NLP Specialization

- Natural Language Processing[#]
- NLP Applications
- Information Retrieval
- Speech Processing
- Conversational AI
- Social Media Analytics
- Large Language Models for Generative AI
- Advanced Reasoning and Planning

[#]At least 3 courses are required including the course marked in

Pool of Electives – Audio and Vision

- Computer Vision[#]
- 3D Computer Vision
- Audio Analysis[#]
- Video Analysis
- Computational Photography
- Computational Imaging
- Contemporary Computer Graphics

[#]At least 3 courses are required including the course marked in

Pool of Electives General

- MLOps
- Data Management for Machine Learning
- Advanced Data Mining
- Design of Algorithms
- AI and ML for Robotics
- AI and ML Techniques for Cyber Security
- Probabilistic Graphical Models
- Metaheuristics for Optimization
- Automated Reasoning
- Quantum Machine Learning
- Software Engineering for Machine Learning
- Introduction to Parallel and Distributed Programming
- API-driven Cloud Native Solutions
- Agentic AI Systems
- Multimodal AI
- Architecting AI Systems
- Multimodal Information Retrieval

Project

- Dissertation



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M.Tech. Artificial Intelligence and Machine Learning

Courses Wise Syllabus

Deep Neural Networks

Introduction to neural networks, linear neural networks for regression and classification, deep feedforward neural networks, backpropagation, deep network training, regularization and optimization, transfer learning and fine-tuning, convolutional neural networks, recurrent neural networks, attention mechanisms, transformers and vision transformers.

Machine Learning

Introduction to Machine Learning, Various kinds of learning, Supervised Learning, Unsupervised Learning; Machine Learning workflow; Bayesian Learning, MAP Hypothesis, Bayes Optimal Classifier, Naive Bayes Classifier; Linear Models for Regression; Linear Models for Classification; Non-Linear models; Decision trees; MDL Principle; Instance Based Learning; Support Vector Machines; Ensemble Learning; Clustering, k-means, GMM; Machine Learning Model Evaluation/Comparison

Computer Vision

Introduction to computer vision and applications, vision fundamentals, image features (edges, corners, texture, keypoint descriptors), CNNs for vision, image classification architectures (ResNet, EfficientNet), object detection (YOLO, R-CNN families), semantic and instance segmentation (U-Net, Mask R-CNN), visual transformers (ViT), image generation (GANs and diffusion models), vision-language models, image captioning (attention-based and transformer models), visual question answering, agentic AI for vision tasks, ethics and privacy in computer vision.

Video Analysis

Introduction to video analysis and applications, video fundamentals (spatial, temporal, motion features, optical flow, spatio-temporal features), temporal and spatio-temporal models (3D CNNs, video transformers), action recognition and temporal action detection, video object detection and tracking, video segmentation, video anomaly detection (including deepfake detection), video captioning and video question answering, video generation (GANs, diffusion models, text-to-video), video summarization, agentic AI for video, ethics and privacy in video analysis.



Audio Analysis

Introduction to audio analysis and applications, audio fundamentals and feature extraction, acoustic AI applications (healthcare monitoring, industrial predictive maintenance, ultrasound applications), deep learning models for audio (CNNs, RNNs/LSTMs, audio transformers), audio classification and tagging, speech-language models (ASR, TTS, voice assistants), music analysis and recommendation, audio-language models (audio captioning, audio question answering), audio generation (GANs, diffusion models, text-to-audio, music generation), agentic AI for audio, ethics and privacy in audio analysis.

Large Language Models for Generative AI

Foundations of Large Language Models, Pre-trained LLMs, Inference using LLM, Fine-tuning techniques, Preference alignment, Prompt Engineering and Reasoning, Scaling and Optimizing LLMs, Retrieval-Augmented and Retrieval-Enhanced Language Models, LLMs as Agents, Structured Data Handling, Evaluation, Challenges, Security and privacy, Advanced Topics

NLP Applications

Sentiment Analysis & Opinion Mining, Machine Translation including Indic Languages, Question answering and Conversational AI, Information extraction, Knowledge graph (RAG) Applications, Security, Privacy & Ethics in NLP, Scalable & Efficient NLP Systems, Implementation of above NLP applications using state of art Generative AI and Agentic AI technologies

Conversational AI

Design and build real-world conversational AI agents that can think, plan, and act autonomously. Develop intelligent systems using Large Language Models, memory, tool integration, and information retrieval. Production-ready skills including system architecture, security, ethical considerations, cost optimization, and performance monitoring. Design, implement, evaluate, and deploy scalable conversational AI agents for enterprise applications.



Social Media Analytics

Social media generates massive amounts of noisy, unstructured, and multimodal data including text, images, audio, and video. This course covers social media data collection, preprocessing, and analysis. It includes social network analysis, influencer detection, sentiment analysis, and trend discovery to understand user behavior and information diffusion. Multimodal analytics and visualization techniques are used to extract actionable insights and support decision-making. The course also addresses security aspects such as fake account, bot, spam, and misinformation detection. Governance topics include data privacy, ethical use, regulatory compliance, and responsible management of social media analytics systems.

AI and ML Techniques for Cyber Security

Introduction to AI and ML in Cybersecurity; Cybersecurity Fundamentals - vulnerabilities, threats, attack vectors, and defense strategies; Vulnerability Management using Gradient Boosting algorithms; Code Security and Vulnerability Detection using graph-based static analysis; Graph Neural Networks for malware call graphs, attack paths, and code vulnerability analysis; Malware Detection and Classification using supervised learning and deep learning; Phishing and Email Security using NLP and Transformer models; Network Security Monitoring using deep learning for traffic analysis; User and Entity Behavior Analytics using autoencoders for anomaly detection; Robustness and Defense using Adversarial Machine Learning; GenAI and LLMs in Security Operations; Automated Security Response using Reinforcement Learning

Agentic AI Systems

Foundations of Agentic AI, Industry use cases of agentic systems, LLMs as Autonomous Agents, Prompt engineering for agents, ReAct, Plan-and-Execute, Reflexion agents, Memory and context management, Knowledge-Grounded Agentic Systems, Multi-Agent Systems and Collaboration, Safety, Governance, and Explainability, Evaluation, Deployment, and MLOps, Build a goal-driven LLM agent with tool calling

Advanced Reasoning and Planning

Introduction to Reasoning and Planning, LLM Reasoning, Large Reasoning Models, Reasoning-oriented Prompting, Inference-Time Scaling, Verification, RL for Reasoning, Knowledge Representation and Memory, Hybrid Reasoning, LLM-based Planning, Multi-LLM Collaboration, Evaluation, Safety, and State of the Art



Multimodal AI

Foundations of Multi-modal ML, Different input modalities, Multimodal Fusion and Fission, Multimodal Models & Pretraining, Cross-Modal Alignment & Grounding, Visual Understanding & Reasoning, Generative Multimodal AI, Temporal Intelligence, Embodied AI, Ethics and safety, Advanced Topics

Architecting AI Systems

Course Description is same as SS* ZG509

Advanced Data Mining

Foundations of Data Mining & Record Data Mining, Association Analysis and Pattern Discovery, Topics beyond conventional record data mining. Mining complex data structures. Tree/graph mining, sequence mining, web/text data mining, stream data mining, spatiotemporal data mining, mining multi-variate time series data, high-dimensional data clustering, and mining social networking sites. Mining data from multiple relations (Multi-relational Data Mining). Privacy preserving Data Mining. Distributed computing solutions for data intensive data mining.



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Assessment Methodology



Continuous Assessment

Quizzes, assignments, mid-semester and comprehensive exam



Case studies & Assignments

Real-world cases used as in-class problem-solving exercises



Dissertation/ Project Work

- Apply learnings to a complex, real-world project
- Demonstrate understanding of learnt concepts

Mode of Examination

- Applicable for students admitted from July 2026 batch onward.
- Semesters 1–3 include mid-semester and comprehensive exams for each course.
- In-person exams mostly held on Friday, Saturday or Sunday at designated centers.
- Quizzes/assignments conducted online via Learning Management System (LMS).
- Semester 4 includes Dissertation/Project Work as per guidelines.



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Exam Centre for Indian Students

Students can take exams at any of our designated centres, including:



Bangalore - Central, Bangalore - East, Bangalore - North, Bangalore - South, Bangalore - West, Belagavi, Chennai - Central, Chennai - East, Chennai - North, Chennai - South, Chennai - West, Coimbatore, Hosur, Hubli-Dharwad, Hyderabad, Hyderabad-Secunderabad, Anantapur, Vijayawada, Visakhapatnam, Rajahmundry, Tirupati, Kochi, Kannur, Kozhikode, Thiruvananthapuram, Thrissur, Madurai, Mangaluru, Mysore, Salem, Pondicherry, Udupi, Vellore.



Delhi NCR - Delhi, Delhi NCR - Faridabad, Delhi NCR - Gurugram, Delhi NCR - Noida, Ghaziabad, Agra, Allahabad (Prayagraj), Alwar, Bathinda, Chandigarh, Dehradun, Jaipur, Jodhpur, Lucknow, Pilani - BITS Campus, Renukoot, Udaipur, Varanasi.



Ahaliyanagar (Ahmednagar), Ahmedabad, Bharuch, Goa, Indore, Bhopal, Jamnagar, Mumbai, Mumbai - Navi Mumbai, Mumbai - Thane, Nagpur, Nashik, Pune, Raipur, Rajkot, Satara, Surat, Vadodara.



Kolkata, Guwahati, Bhubaneswar, Jamshedpur, Patna, Ranchi, Silvassa.

Exam Centre for International Students

- A designated international exam centre is available in Dubai, Abu Dhabi (UAE), Doha (Qatar), Kuala Lumpur (Malaysia), Singapore, Riyadh (Saudi Arabia) and Muscat (Oman).
- Students outside Dubai, Abu Dhabi (UAE), Doha (Qatar), Kuala Lumpur (Malaysia), Singapore, Riyadh (Saudi Arabia) and Muscat (Oman) can opt for online exams by submitting:
 - a) Scanned copy of valid visa for the country in which you are currently residing.
 - b) Scanned copy of government-issued ID from the residing country.
 - c) HR recommendation or endorsement letter from the employer, stating the location of your work.
- Indian students temporarily based out of India can also avail online exams on request with the above-mentioned documents.



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Fee Structure

The following fees schedule is applicable for candidates seeking new admission during the academic year 2026-27.



Easy Monthly Payment Option with 0% Interest and 0 Down Payment

Instant EMI option with 0% interest and 0 Down Payment is now available that allows you to pay programme fee in an easy and convenient way.

- Instant online approval in seconds
- No Credit Cards/ CIBIL score required
- Easy & Secure online process using Aadhaar and PAN number
- Anyone with a Salary Account with Netbanking can apply Option to submit fee using easy - EMI with 0% interest and 0 down payment

Admissions Open for July 2026 Batch

All the above fees are non-refundable.

Important: For every course in the programme, institute will recommend textbooks, students would need to procure these textbooks on their own.

[Click here](#)

to learn more

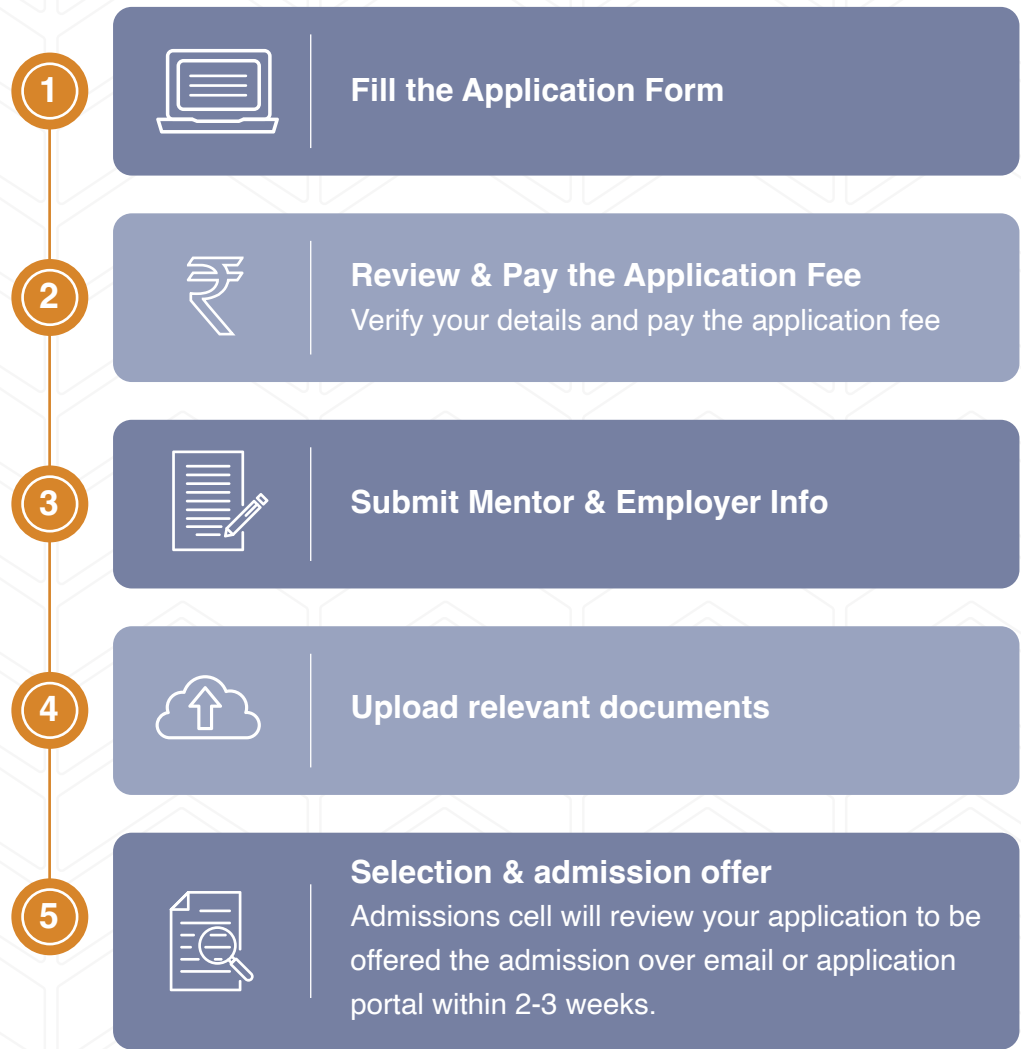


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How to Apply

Login to our Application Centre to apply for the programme. [Click Here](#).



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Students Speak



BITS Pilani WILP provided me with the opportunity to pursue a structured Master's programme without me having to leave my job. I was able to enhance my expertise in Python, Machine Learning, NLP, and Deep Learning, enabling me to successfully transition from a business consultant to a project engineer specializing in AI. Further, the faculty at BITS Pilani WILP were outstanding and had rich industry experience. The flexibility of online lectures and access to recorded sessions enabled me to effectively balance my professional, personal, and academic commitments.

Anand Jha

Lead Data Engineer - AI, Intellect Design Alumnus



BITS Pilani WILP enabled me to enhance my skills while continuing to work, providing a comprehensive understanding of Data Mining, Machine Learning, AI, and Big Data Systems, which significantly boosted my problem-solving and analytical abilities. The hands-on tools and remote labs bridged the gap between theory and practice, allowing me to confidently apply advanced technologies to real-world challenges. For my final semester project, I focused on cross-selling products by analyzing customer data and behavior, integrating data mining and machine learning techniques to turn theoretical concepts into actionable insights and showcasing the programme's real-world relevance.

Nikhila

Senior Data Analyst, Citi Alumnus



BITS Pilani WILP played a key role in my transition from an SAP developer to an AI architect, providing me with industry-relevant skills that enabled me to handle complex projects and contribute at a higher level within my organization, all while seamlessly balancing work and academics. The programme's focus on continuous learning expanded my horizons, giving me the confidence to explore new opportunities. The combination of theoretical knowledge, practical tools, and strong mentor support made this journey a key milestone in both my personal and professional development.

Manasa

AI Architect Alumnus



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Workforce Transformation Since 1979

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